

Alice (initiator)

Bob (responder)

Publishes Extended Pre-Key Bundle

- IK_B , $SPK_B + \text{Sig}_{IK_B}(SPK_B)$, $\{OPK_B\}$,
- $PQPK_B + \text{Sig}_{IK_B}(PQPK_B)$ (KEM public key)

Fetches Extended Pre-Key Bundle

Alice Computes:

- Verify $\text{Sig}(SPK_B)$ and $\text{Sig}(PQPK_B)$
- Generate Ephemeral Key Pair EK_A
- $(CT, SS) \leftarrow \text{Encaps}(PQPK_B)$
- $DH1 \dots DH4$ (as in X3DH)
- **$SK = \text{KDF}(DH1 \parallel DH2 \parallel DH3 \parallel DH4 \parallel SS)$**

Initial Message: IK_A , EK_A , Identifiers of OPK_B and $PQPK_B$, CT

Bob Computes

- $SS \leftarrow \text{Decaps}(sk_{PQPK_B}, CT)$
- Recompute $DH1 \dots DH4$
- **$SK = \text{KDF}(DH1 \parallel DH2 \parallel DH3 \parallel DH4 \parallel SS)$**
 - (same SK as Alice)

Hybrid Construction: SK depends on both the Curve25519 values $DH1$ – $DH4$ and the KEM secret SS , so recovering SK requires defeating both the discrete-logarithm problem and the KEM. The output SK is handed to the Double Ratchet exactly as X3DH's output was.